

RESEARCH INTERESTS

Strong gravitational lensing, sub-galactic scale dark matter, near-infrared space telescopes

EDUCATION

Washington University in St. Louis Ph.D. in Physics – Advisor: Dr. Tansu Daylan – A.M. in Physics (en route) achieved May 2024	St. Louis, MO, USA 2022–Present
Trinity College, University of Cambridge M.Phil. in History and Philosophy of Science and Medicine, <i>First Class Honours</i> – Dissertation: “Wavefunction ontology and Koopman-von Neumann theory” – Supervisor: Dr. Jeremy Butterfield	Cambridge, UK 2018–2019
Kenyon College B.A. in Physics, Philosophy, <i>magna cum laude</i>	Gambier, OH, USA 2014–2018
Lady Margaret Hall, University of Oxford Visiting Student Programme, Physics and Philosophy	Oxford, UK 2016–2017

PROFESSIONAL EXPERIENCE

- Technical Consultant**, R&D Technical Services, Veeva Systems Oct 2021–Jun 2022
Developed and implemented custom software integrations with Veeva Vault and extended system functionality utilizing Vault Java SDK and REST API, Amazon Web Services. Managed integration and development projects.
- Associate Consultant**, R&D Services, Veeva Systems Oct 2019–Oct 2021
Implemented Veeva’s Vault cloud software applications at global life sciences companies. Led candidate case study interviews and mentored new hires’ mock projects.

HONORS AND AWARDS

- 1st Place, Student E-Poster Competition (Physical Sciences category)** 2026
2026 Annual Meeting of the American Association for the Advancement of Science
- 1st Place, Poster Competition**, Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era 2026
- 1st Place (Graduate Division)**, Washington University Physics Research Symposium 2025
- 1st Place (Graduate Division)**, Washington University Physics Research Symposium 2024
- Future Investigators in NASA Earth and Space Science and Technology (FINESST)** 2024
Maximizing JWST and Roman Dark Matter Science with Strong Gravitational Lensing
\$150,000 award over three years
- First Class (Distinction)**, M.Phil. Dissertation, University of Cambridge 2019
- Member, Phi Beta Kappa**, Beta of Ohio, Kenyon College 2018
- Distinction**, Senior Exercise in Physics, Department of Physics, Kenyon College 2018
- Distinction**, Senior Exercise in Philosophy, Department of Philosophy, Kenyon College 2018

- **Member, Sigma Pi Sigma**, the Physics Honor Society (American Institute of Physics) 2017
- **Merit List Scholar**, Kenyon College 2014–2018
- **Trustee Opportunity Scholarship**, Kenyon College 2014–2018

PUBLICATIONS

- [1] N. Khadka, S. Birrer, H. Best, P. Sharma, K. T. Abe, X. Tang, C. Mistick, F. Urcelay, E. M. Sonmez, N. Arendse, S. Erickson, J. O. Hjortlund, P. Holloway, A. Huang, R. Karthik, M. Lamontagne, V. Negi, J. R. Pierel, B. Sanchez, A. Ece Saricaoglu, A. Shajib, Y. Shao, P. Venkatraman, **B. Wedig**, A. Agrawal, T. Anguita, P. Bessa, C. R. Bom, S. Castillo, T. Collett, T. Daylan, S. Dillmann, M. Grespan, E. E. Hayes, R. Joseph, R. Kessler, T. Li, P. Marshall, A. More, V. Motta, G. Narayan, M. O'Dowd, M. Oguri, A. Verma, G. Vernardos, the Strong Lensing Science Collaboration, and the LSST Dark Energy Science Collaboration, “SLSim: a strong lensing population simulation package”, *arXiv e-prints*, arXiv:2603.17138, arXiv:2603.17138, Mar. 2026. arXiv: 2603.17138 [[astro-ph.CO](#)].
- [2] **B. Wedig**, T. Daylan, S. Birrer, F.-Y. Cyr-Racine, C. Dvorkin, D. P. Finkbeiner, A. Huang, X. Huang, R. Karthik, N. Khadka, P. Natarajan, A. M. Nierenberg, A. H. G. Peter, J. D. R. Pierel, X. T. Tang, and R. H. Wechsler, “The Roman View of Strong Gravitational Lenses”, *The Astrophysical Journal*, vol. 986, no. 1, 42, p. 42, Jun. 2025. arXiv: 2506.03390 [[astro-ph.CO](#)].

PRESENTATIONS

Invited Talks

- “The Roman View of Strong Gravitational Lenses” Dec 2025
“Nature of Dark Matter at Small Scales” Virtual Seminar Series

Contributed Talks

- “Probing Dark Matter Microphysics with JWST’s Galaxy-Galaxy Strong Gravitational Lenses” Dec 2025
53rd Mid-American Regional Astrophysics Conference, Columbia, MO, USA
- “A JWST Survey of Subhalos Across Cosmic Time” Oct 2025
1st Workshop Scaling up Lensing, University in Liège, Liège, Belgium
- “The Microphysics of Dark Matter with the Habitable Worlds Observatory” Jul 2025
Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology Washington, D.C., USA
- “Hunting for Dark Matter with NASA’s Next Space Telescope” Apr 2025
2025 Graduate Research Symposium, Washington University in St. Louis
- “The Roman View of Strong Gravitational Lenses” Jan 2025
245th Meeting of the American Astronomical Society, National Harbor, MD, USA
- “The Roman View of Strong Gravitational Lenses” Dec 2024
52nd Mid-American Regional Astrophysics Conference, Lawrence, KS, USA
- “The Roman View of Strong Gravitational Lenses” Jul 2024
Challenging Theory with Roman: From Planet Formation to Cosmology, Pasadena, CA, USA
- “Image Simulation of Strong Gravitational Lenses Detectable by the Roman Space Telescope” Jan 2024
243rd Meeting of the American Astronomical Society, New Orleans, LA, USA

Seminars

- “Searching for dark matter substructure in JWST’s galaxy-scale strong lenses” Nov 2025
Washington University Cosmology and Astroparticle Physics Meeting
Department of Physics, Washington University in St. Louis
- “Hunting for Dark Matter with Space Telescopes” Feb 2025
A&S Research Roundtable, School of Arts & Sciences, Washington University in St. Louis
- “Maximizing JWST and Roman Dark Matter Science with Strong Gravitational Lensing” Nov 2024
Washington University Cosmology and Astroparticle Physics Meeting
Department of Physics, Washington University in St. Louis
- “The Roman View of Strong Gravitational Lenses” Nov 2024
Physics Graduate Student Seminar, Department of Physics, Washington University in St. Louis
- “The Roman View of Strong Gravitational Lenses” Apr 2024
Washington University Cosmology and Astroparticle Physics Meeting
Department of Physics, Washington University in St. Louis
- “Investigating Dark Matter Substructure with Gravitational Lensing” Nov 2023
Physics Graduate Student Seminar, Department of Physics, Washington University in St. Louis
- “Lessons for wavefunction ontology from Koopman-von Neumann theory” Easter 2019
MPhil Seminar, Department of History and Philosophy of Science, University of Cambridge
- “Assessing the Quantum Liar Paradox” Lent 2019
MPhil Seminar, Department of History and Philosophy of Science, University of Cambridge
- “Is Nature Nonlocal?” Apr 2018
Senior Exercise, Department of Physics, Kenyon College
- “Reducing quantum noise in LIGO: Characterization of an ultra-low loss polarizing beam splitter” Aug 2016
Ole Miss Chemistry Summer Research Program, University of Mississippi

Poster Presentations

- “Forecasting Dark Matter Subhalo Detections with NASA’s Habitable Worlds Observatory” Feb 2026
2026 Annual Meeting of the American Association for the Advancement of Science
Phoenix, AZ
- “Forecasting Dark Matter Subhalo Detections with NASA’s Habitable Worlds Observatory” Nov 2025
Washington University Physics Research Symposium
Department of Physics, Washington University in St. Louis
- “Hunting for Dark Matter with NASA’s Next Space Telescope” Apr 2025
50th Anniversary Symposium, McDonnell Center for the Space Sciences
Washington University in St. Louis
- “Hunting for Dark Matter with NASA’s Next Space Telescope” Apr 2025
2025 Imaging Science Pathway Retreat, Washington University in St. Louis
- “The Roman View of Strong Gravitational Lenses” Nov 2024
Washington University Physics Research Symposium
Department of Physics, Washington University in St. Louis
- “Image Simulation of Strong Gravitational Lenses Detectable by the Roman Space Telescope” Nov 2023
Washington University Physics Research Symposium
Department of Physics, Washington University in St. Louis

TEACHING

- **Guest Lecturer**, Astrostatistics (PHYS 4680/5680) Spring 2025
Department of Physics, Washington University in St. Louis
- **Guest Lecturer**, Ampersand: Gateway Expeditions into Exoplanets (PHYS 1210) Fall 2024
Department of Physics, Washington University in St. Louis
- **Assistant in Instruction**, Physics I Laboratory (PHYS 191L) Fall 2023
Department of Physics, Washington University in St. Louis
- **Assistant in Instruction**, Physics II Laboratory (PHYS 192L) Spring 2023
Department of Physics, Washington University in St. Louis
- **Apprentice Teacher**, Intensive Introductory Japanese (JAPN 111-112Y) Fall 2017
Department of Modern Languages and Literatures, Kenyon College
- **Teaching Assistant**, Advanced Topics in Physics: Special Relativity (SREL) Summer 2017
Johns Hopkins Center for Talented Youth
- **Tutor**, Oscillations and Waves (PHYS 245) Spring 2016
Department of Physics, Kenyon College
- **Apprentice Teacher**, Intermediate Japanese (JAPN 213-214Y) Fall 2015–Spring 2016
Department of Modern Languages and Literatures, Kenyon College
- **Apprentice Teacher**, Intensive Introductory Japanese (JAPN 111-112Y) Fall 2014–Spring 2015
Department of Modern Languages and Literatures, Kenyon College

MENTORING

- **Jodie Xiao**, Undergraduate student at Washington University in St. Louis May 2025–Present
Participated in the LSST Strong Lens Data Challenge in fall 2025. Trained a convolutional neural network to classify strong lenses, achieved score 98.1/100 placing 9th out of 21 teams. Presented results at Fall 2025 WashU Undergraduate Research Symposium and 2025 Washington University Physics Research Symposium.

SERVICE AND PUBLIC ENGAGEMENT

- **Speaker**, Washington University Postbaccalaureate Research Education Program Feb 2026
“Introduction to Science Policy/Advocacy”
- **Judge**, Chambliss Astronomy Achievement Student Awards Jan 2026
247th Meeting of the American Astronomical Society
- **Speaker**, “From Variable Stars to Dark Matter – The Expanding Legacy of Henrietta Leavitt” Nov 2025
WashU Departments of Physics & Performing Arts
- **Guest Author**, Astrobites Jun 2025
“The Surprising Discovery of a Distant, Nearly Perfect Syzygy”
- **Exam Author**, 38th Annual High School Physics Contest Apr 2025
St. Louis Area Physics Teachers
- **Participant**, Catalyzing Advocacy in Science and Engineering Workshop Apr 2025
American Association for the Advancement of Science
- **Participant**, Congressional Visits Days Apr 2025
American Astronomical Society
- **Judge**, Chambliss Astronomy Achievement Student Awards Jan 2025
245th Meeting of the American Astronomical Society
- **Co-organizer**, Splitter Session Jan 2025

“ExoCore: An open science curriculum for enhanced reproducibility and equity in exoplanet research”
245th Meeting of the American Astronomical Society

- **Contributor**, “ExoCore: An Open Science Curriculum for Exoplanet Research” Dec 2024–Present
NASA Open Science Initiative
- **Speaker**, Astronomy on Tap STL Oct 2024
“Hunting for Dark Matter with NASA’s Next Space Telescope”
- **Co-president**, Washington University ProSPER 2024–Present
Science communication and policy graduate student group
- **Exam Author**, 37th Annual High School Physics Contest Apr 2024
St. Louis Area Physics Teachers

Department of Physics, Washington University in St. Louis

- **Graduate Student Head Mentor** 2025–2026
- **Speaker**, Saturday Science Lecture: “Seeing the dark with gravitational lensing” Mar 2025
- **Co-organizer**, Washington University Cosmology and Astrophysics Seminar 2024–2025
- **Graduate Student Mentor** 2023–Present
- **Member**, Computing Committee 2023–Present
- **Member**, Spaces Committee 2023–2025
- **Member**, BBQ Committee 2023–2025

WORKSHOPS AND SCHOOLS

- Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era Jan 2026
University of Arizona
- JWST Summer School on High-Redshift Transients Aug 2025
Space Telescope Science Institute
- ALMA Cycle 12 Proposal Preparation Workshop Apr 2025
University of Missouri